



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 13.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BITS F225
Course Title : Environmental Studies
Instructor-in-Charge : B.Vani (Biological Sciences)
Instructor(s) : Prof. Banasri Roy, Ms Varsha (Chemical), Dr. Ravi Kumar Guntu (Civil).
Room No. : 5102

1. Course Description: Environment, human population, and industrialization; natural resources and the impact of man-made activities on them; structure and function of ecosystem, population ecology, biodiversity and its conservation, environmental pollution, social issues and the environment, and environmental impact assessment.

2. Scope and Objective of the Course: The United Nations Conference on Environment and Development (UNCED) held in Rio De Janero in 1992, the 1994 conference on United Nations Framework Convention on Climate Change (UNFCCC), and a very recent 2015 Paris Agreement have created awareness among the people around the globe about the deteriorating condition of Earth's environment. It has warned us not to be ignorant of the environmental issues. In lieu of the situation, the Hon'ble Supreme Court directed the UGC to introduce a basic course on environment for undergraduate students in India.

The course is designed to cover all aspects of environment [Ecological including biotic (flora- fauna) and abiotic (air, water, soil & geography), social, economic and aesthetical]. The course is of interdisciplinary nature dealing with social, biological & engineering aspects of environment. The role of biotic and abiotic factors in the functioning of the environment, the impact of anthropogenic activities on the environment, different environmental pollutants, its impact and management strategies; and ecological, socio-political and economic issues will be discussed with appropriate real-life examples and case studies. The course would also provide insight into practical solutions to different environmental issues. Finally, important concepts of environmental impact assessment from all the above mentioned sources would be dealt with. The interested students can take the learnings forward from this point and enrich themselves further by doing other electives/projects in these areas.

3. Text Books: Erach Bharucha, Text Book of Environmental Studies for Undergraduate Courses, University Press (India) Private Limited, Second Edition, 2005.

4. Reference Books:

1. Richard T. Wright, Dorothy F. Boorse, Gordon College, Environmental Science: Toward a Sustainable Future, Pearson Publishers, 12th Edition, 2014.
2. Gerard Kiely, Environmental Engineering, McGraw Hill Education, Special Indian Edition, 2007.
3. Gilbert M. Masters and Wendell P. Ella, Introduction to Environmental Engineering and Sciences, PHI Learning Private Lt d, 3rd Edition, 2007.
4. Dynamically shared on Google class room



5. Course Plan:

Note: The entire course is divided into multiple streams [Bio, Civil, and Chemical].

Introductory Lecture (All departments) – 1 Lecture			
Introduction to Environmental Studies	L0: Introduction to the course and importance of Environment.	<i>T1-Unit 1 & Class notes</i>	Understand the modality of the course, environmental issues and its impact of life on Earth

For Bio-Sciences: Total 8 Lectures

Module	Lecture session	Reference	Learning Outcome
Concept of ecosystems	L1: Structure of Ecosystems, Global biomes and Climate. L2: Energy and Material flow in ecosystems	<i>T1-Unit 3; R1 Ch-5 and 7 & Class notes</i>	Learn about structure and functioning of ecosystems, biomes and climate
Biodiversity and its conservation	L3, L4: Biodiversity, its value and measurement methods	<i>T1-Unit 4; R1 Ch-6 & Class notes</i>	Know about the biological wealth and its value. Learn biodiversity estimation methods, understand why there is biodiversity crisis and methods/strategies for its conservation
Biodiversity and its conservation	L5, L6 : Biodiversity crisis and conservation strategies	<i>T1-Unit 4; R1 Ch-6 & Class notes</i>	Know about the biological wealth and its value. Learn biodiversity estimation methods, understand why there is biodiversity crisis and methods/strategies for its conservation
Population Ecology	L7: Dynamics of natural populations L8: Mechanisms of Population Equilibrium, Response of population to environment	<i>T1-Unit 7; R1 Ch-4, 8 and 9 & Class notes</i>	Learn about different growth curves, intra and inter-species interactions. Understand evolution as a force for change, ecosystem responses to disturbance, ecological succession

For Civil Engineering: [13 Lectures]

Module	Lecture session	Reference	Learning Outcome
Need and importance of water resources	L1 and L2: Water Cycle, Water Requirements, Water resources, utilization and its	<i>T1-Unit 2 & Class Notes</i>	Present day status/over utilization of water resources and their pollution. Water stressed countries, disputes and conflicts over water



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	management, Smart cities and water requirements		Need for Water Resource Management
Development and Regulation of Water Resources	L3 & L4: Water Resource Regulations in India	<i>T1-Unit 6 & Class notes</i>	Rain water harvesting, Innovations in irrigation techniques. Engineering considerations for construction of dams and major infrastructure projects
Disaster mitigation and management	L5, L6: Types of Natural Disaster, Hazard Identification, Risk Analysis and Management	<i>Class Notes and PPT</i>	Understand the importance of disaster mitigation, identification of various disasters and preparedness for their effective management
Building and Environment, Smart Cities	L7-L10: Building construction and its impact on environment; Concept of smart cities; Innovations in Building construction; Construction Planning	<i>Class Notes and PPT</i>	Understand the importance of holistic building construction, concept of smart cities etc.,
Air Pollution	L11: Criteria pollutants, Non criteria pollutants, L12: Air pollution meteorology, Atmospheric Dispersion	<i>R2 – Ch 8, 8.1, 8.3, 8.4, 8.6 - 8.10</i>	Gives an understanding of different types of air pollutants, effect of atmosphere on pollutants behavior, specific case study of carbon capture sequestration

Chemical: [Total Lectures: 13 Lectures]

Module	Lecture session	Reference	Learning Outcome
Environmental Pollution	L1, L2 : Pollution: Air, Water, Soil, Marine, Noise, Thermal, Nuclear & Related Aspects	<i>T1 –Unit 5, 5.1 – 5.8& Class notes</i>	An overview of environmental pollution, which will unfold as details are covered
Environmental Laws and Regulations-	L3 and L4: Legal framework of Environmental Laws & regulations	<i>R2 – Ch 1, 1.2 – 1.3, 1.6, 1.8 & class notes</i>	Gives an over view of US-EPA laws and CPCB laws & standards followed
Technologies employed for industrial air emissions control	L5: Characterizing the Air Stream, Equipment Selection, Different Types of Air Pollution Control Techniques	<i>R2 – Ch- 16 and Class notes</i>	Characterization of air stream(s), so that suitable equipment (technology) can be selected for industrial air emissions control



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Design aspects of three major technologies for industrial air emissions control	L6, L7, L8: Equipment Design: Scrubbers Absorption Adsorption	<i>R2 – Ch - 16 and Class notes</i>	Appreciation and understanding of the principles (involved) which help in designing the equipment for the control of industrial air emission(s)
Renewable and Clean energy Technologies	L9-10 Solar, Wind, geothermal, and Hydrogen energy technologies	<i>Lecture Notes</i>	Understand the concepts of alternative clean energy technologies and their importance for sustainable living
Sanitation science and engineering	L11 and L12: Sustainable Development Goal 6 (SDG 6) and alarming urgency of ‘on the ground’ professional impact on clean water and sanitation	Reference Notes	Appreciation of sanitation science and engineering w.r.t SDG and understanding of technological developments in the field of sanitation

6. Evaluation Scheme:

S.No.	Component	Duration	Weightage (100 %) 200 Marks)	Date & Time	Remarks
1.	Mid Semester Test	0.5 hr.	35 % (70 M)	(50%Che+50% Bio)	CB 4/10/2026 IPC
2.	Comprehensive Examination	1.0 hr.	45 % (90 M)	(Bio20%+Che 20%+60% Civil)	CB 30/11/2026 FN IPC
3.	Quiz (Announced)	20 min.	20 % (40 M)	(Bio30%+Civil 30%+ Che40%)	Closed Book TBA

7. Online Consultation Hour: To be announced in the class.

8. Notices: All Notices would be put up on Nalanda.

9. Make-up Policy: Make-up is granted only for genuine cases with valid justification and prior permission of Instructor-in-charge. Average marks of both quizzes will be considered. No makeup for quizzes.

Instructor-in-charge
BITS F225